

**TECHNICAL SPECIFICATION FOR HIGH TENSION LINE HARDWARE AND
ACCESSORIES FOR ACSR WOLF CONDUCTOR**

1.1.0 SCOPE :

This specification covers design, manufacture, testing at manufacturer's works before despatch, supply and delivery of Conductor Accessories and Line Hardware for Over Head Lines to be used as 33KV Line by Wolf Conductor.

1.2.1 STANDARDS:

The materials covered under this specification shall comply with the requirements of the latest version of the following standards as amended upto date, except where specified otherwise.

i)	IS-2486 Part-I,II & III :	Insulator fittings for Over Head Power Lines with nominal voltage greater than 1,000 Volt.
ii)	IS-2121 Part-I & II :	Conductor Accessories for Over Head Power Lines
iii)	IS-2629 :	Recommended practice for hot dip galvanizing of Iron and Steel.
iv)	IS-2633 :	Method of testing for uniformity of coating on Zinc coated articles.

- 1.2.2 The materials conforming to any other national standards which ensure equal or better performance shall also be acceptable. The salient point of these specifications and points of difference between these and the above specifications, shall be clearly brought out in the Bid.

1.3.0 MATERIAL & WORKMANSHIP :

- 1.3.1 The material offered shall be of best quality, workmanship, well finished and of latest proven design and conform to the best modern practices adopted in the extra high voltage field. All castings shall be free from internal defects like blow-holes, cracks, inclusion, shrinkage etc. and shall be smooth. All mechanical surface shall be true, smooth and well finished.
- 1.3.2 Metal fittings of specified material for Conductor Accessories and overhead Line Hardware are required to have excellent mechanical properties such as strength, toughness and high corrosion resistance and free from corona formation. All current carrying parts shall be so designed and manufactured that contact resistance is reduced to the minimum and localized heating phenomenon is averted.
- 1.3.3 All items shall be supplied complete in all respect and shall include all fittings necessary for completing the assembly. The tenderers shall be responsible for satisfying himself that the Accessories and Hardware offered are complete and suitable for the proposed attachment and for the size of the specified conductor.
- 1.3.4 All bolts, nuts, bolt-heads shall be the whitworth standard thread. Bolt heads and nuts shall be hexagonal. However, if required, nuts shall be locked in an approved manner. The thread in nuts and tapped holes shall be cut after galvanizing and shall be well lubricated or greased. All other threads shall be cut before galvanizing. The bolt threads shall be under-cut to take care of increase in diameter due to galvanizing. Rivet (Cotterpin) in place of Bolt & Nuts shall not be acceptable. Bolts & nuts should be ASP/BI/GKW or equivalent make. Spring washers shall be electro galvanized.
- 1.3.5 The general design of Conductor Accessories and overhead Line Hardware shall be such as to ensure uniformity, high strength, freedom from corona formation and high resistance against corrosion even in case of high level of atmosphere pollution.
- 1.3.6 All hooks, eyes, pins, bolts, Suspension Clamps and other fittings for attaching insulators to the Rail Pole or to the line conductor shall be so designed as to reduce to a maximum any damage to the conductor and insulator or other fitting arising from the conductor vibration.
- 1.3.7 The design of the fittings shall be such as to avoid local corona formation or discharges likely to cause interference to tele-Transmission of signals of any kind.

1.4.0 GALVANIZATION :

- 1.4.1 All ferrous parts of Conductor Accessories and Line Hardware shall be galvanized in accordance with IS: 2629-1966 recommended practice for hot dip galvanizing of iron and steel or any other equivalent authoritative standards. The weight of zinc coating shall be determined as per methods for testing weights, thickness and uniformity of coating of hot-dip galvanized articles as stipulated in IS: 2633-1972 or any other equivalent authoritative standard. The zinc used for galvanization shall conform to grade ZN 98 of IS:209. The galvanized parts shall withstand four (4) dips of 1 minute each while testing uniformity of zinc coating as per IS: 2633-1972.
- 1.4.2 Spring washers shall be made of spring steel of coil rectangular shape as per IS: 1573 Service Condition 4.

1.5.0 REQUIREMENT OF INSULATOR STRING ASSEMBLIES :

- 1.5.1 Hardware for suspension and tension insulators shall be suitable for insulators with nominal pin shank diameters of 16mm.
- 1.5.2 Each insulator string hardware shall generally include the following hardware components and each set of hardware shall be supplied complete in every respect.
- 1.5.3 Both suspension and tension type of hardware shall be of Ball and Socket type and should invariably be with security clip.
- 1.5.4 Suspension assembly for Single Suspension Set shall comprise of the following:-
- a) Ball Hook
 - b) Socket eye with 'R' type security clip
 - c) Suspension Clamp
- 1.5.5 Tension assembly for Single Tension Set shall comprise of the following:-
- a) Cross Arm Strap
 - b) Ball Clevis
 - c) Socket clevis with 'R' type security clip
 - d) Compression type Dead End Clamp
- 1.5.6 The Suspension and Tension Clamps shall be suitable for ACSR WOLF conductor. Ultimate Mechanical strength of the hardware would in no case be lower than the following :

Conductor Size	Overall Diameter (approx.) (mm.)	Ultimate mechanical strength (Single)	
		Suspension (KN.)	Tension (KN.)
WOLF (ACSR)	18.13	70	70

1.5.7 MATERIALS :

Materials for the various components of the hardwares shall be as follows:-

a) Cross Arm Strap	Mild Steel
b) Ball Hook	Forged steel
c) Ball Clevis	Forged steel
d) Socket Eye/Socket Clevis	Forged steel

e) Suspension Clamp	Aluminium Alloy
f) Dead End Clamp (Compression Type)	
i) Aluminium Sleeve	Aluminium
ii) Steel Sleeve	Forged steel

The bidder shall be responsible to satisfy himself that all the hardware fittings included in a string are entirely suitable for Rail Pole structures and ACSR Wolf conductor.

1.6.0 INSULATOR HARDWARE FITTINGS :

1.6.1 CROSS ARM STRAP:

Cross Arm Strap shall be made of Mild Steel complete with Bolt & Nut.

1.6.2 BALL HOOK:

Ball Hook shall be made from drop forged steel and smooth finished.

1.6.3 BALL CLEVIS/SOCKET EYE/SOCKET CLEVIS :

All these fittings shall be made of forged steel. Balls and sockets shall be suitable for insulators with nominal pin shank diameter of 16mm.

1.6.4 SUSPENSION CLAMP :

Suspension Clamp will consist of one set Clamp made of Aluminium Alloy, one socket eye made of forged steel and one Ball Hook made of forged Steel. Suspension Clamp shall be suitable for ACSR “Wolf” conductor without armour rod. Ultimate Tensile Strength of Hardware shall be 70 KN and Ultimate Tensile Strength of Suspension Clamp shall be 45 KN in case of “Wolf” conductor.

1.6.5 DEAD END CLAMP (COMPRESSION TYPE) :

The dead end clamps shall be suitable for supporting ACSR Wolf Conductor. The steel sleeve of dead end clamp shall be single piece obtained by process of drop forging and shall not be made by joining, welding, shrink fitting or any other process for more than one piece of material. The steel sleeve shall be hot dip galvanized and outer sleeve shall be made of aluminum.

The dead end clamp shall be compression type with provision for compressing jumper terminal at one end. The clamp shall not permit slipping or damage to or failure of the complete conductor at any part thereof at a load less than 95% of the ultimate tensile strength of conductor. The conductivity of the clamp shall not be less than that of conductor. Non-Compression zone shall be clearly marked on each dead end clamp.

1.7.0 UNIVERSAL JOINTING COMPOUND :

PENTEX “HV” Universal Jointing Compound is a chemically inert compound based on a special petroleum to provide high electrical efficiency. Compound provides low resistance connections over the entire operating temperature range. It shall be used with all compression, accessories and bolted fittings.

1.8.0 TESTS AND TEST CERTIFICATES :

A copy of Type Test Report carried out within last 5 years on the tendered item in a NABL accredited Test House/Laboratory shall be submitted along with the tender. Otherwise, the tender may be rejected.

The following tests shall be carried out as per latest amendment of IS: 2486 (Part-I):

A. TYPE TEST FOR CLAMP :

- a) Visual Examination.
- b) Verification of dimensions.
- c) Slip Strength Tests.
- d) Ultimate Strength Tests.
- e) Galvanizing/Electroplating Tests.

B. TYPE TEST FOR INSULATOR STRING FITTINGS EXCEPT CLAMP :

- a) Visual Examination.
- b) Verification of dimensions.
- c) Mechanical Test.
- d) Galvanizing Test.
- e) Chemical Composition Test.

C. ACCEPTANCE TEST FOR CLAMP :

- a) Visual Examination.
- b) Chemical Composition Test.
- c) Verification of dimensions.
- d) Mechanical Tests.
- e) Ultimate Strength Tests.
- f) Galvanizing Tests.

D. ACCEPTANCE TEST FOR INSULATOR STRING FITTINGS EXCEPT CLAMP :

- a) Visual Examination.
- b) Chemical Composition Test.
- c) Verification of dimensions.
- d) Ultimate Strength Tests.
- e) Galvanizing Tests.

E. ROUTINE TESTS FOR BOTH CLAMP & FOR INSULATOR STRING FITTINGS

- a) Visual Examination
- b) Mechanical routine test

1.9.0 SPECIFIC TECHNICAL PARTICULARS :

The specific technical particulars of ACSR “Wolf” Conductor are given as hereunder :

1.9.1 SYSTEM PARTICULARS :

(i) CONDUCTOR :

a) Type	ACSR “WOLF”
b) Stranding & wire Diameter(mm.)	30/2.59 Aluminium
	7/2.59 Steel
c) Approx. total wt. (Kg./Km.)	726
d) Overall Diameter (mm.)	18.13
e) Ultimate Strength (Kg.)	6867

ii) **DISC INSULATORS** :

		SUSPENSION	TENSION
a) Type	:	Ball & Socket	Ball & Socket
b) Ball Size-As per IS:2486 Pt-II	:	16 mm. 'B'	16 mm. 'B'
c) Diameter	:	254mm.	254mm
d) Spacing	:	145mm.	145mm
e) E.M. Strength	:	70KN	70KN
f) Number	:	3 nos	3 nos

iii) **SUSPENSION CLAMPS** :

a) Material	:	Aluminium Alloy & Steel
b) Slip Strength	:	8 to 15% of ultimate strength of conductor
c) Minimum Failing Load (KN)	:	70 KN

iv) **TENSION CLAMPS (COMPRESSION TYPE)** :

a) Material	:	Aluminium & Steel
b) Slip Strength	:	Not less than 95% of ultimate strength of conductor
c) Minimum Failing Load (KN)	:	70 KN

1.10.0 DRAWING:

Drawing shall be according to the Tender document.

1.10.1 Documents to be submitted at the time of physical delivery at consignee stores :

The following documents to be submitted by the vendors to the consignee Stores at the time of despatch to stores by the vendors:-

- a) Copy of purchase Order.
- b) Copy of Despatch Instruction
- c) Inspection Test Certificate
- d) Guarantee Certificate
- e) Proforma Invoice
- f) Calculation Sheet for price Variation on the basis of IEEMA or CACMAI as applicable with base date of order.
- g) Seal list and packing list
- h) Challan in triplicate
- i) Way bill, if applicable.

GUARANTEED TECHNICAL PARTICULARS

Single Suspension Hardware Fittings for ACSR Wolf Conductor.

1. INSULATOR STRING HARDWARE :

i) Maker's Name and Address :

ii) Size and designation of ball and socket with
Standard specification to which conforming:

iii) Material :

a) Ball Hook	
b) Socket Eye	
c) Suspension Clamp	

iv) Minimum Failing Load

v) Standard specification to which galvanizing conforms

vi) Reference to drawing attached : Drawing shall be according to the Tender document.

2. SUSPENSION CLAMPS :

i)	Material	
ii)	Breaking Strength	
iii)	Slipping Strength	
iv)	Make of Bolts and nuts used	

Signature

Name

Seal of the Firm

Name of the Firm

DRAWING

Single Suspension Hardware Fittings for ACSR Wolf Conductor.

